TEXORA · LECTURE SHEET

Heat Treatment, Moulding and Casting

Two ways of giving metal its final character: heat treatment, which re-engineers steel from the inside by heating and controlled cooling — and casting, which gives metal its shape in one pour, through patterns, moulding sand and a gating system that feeds the mould.

ME-1208 · ENGINEERING WORKSHOP · TOPIC 5 · LEVEL 1 TERM 2

Part A — Heat treatment of steel

1. Why heating and cooling change steel

Steel's properties are set not just by its chemistry but by its **microstructure** — the arrangement of its phases. At room temperature a plain carbon steel is a mixture of soft **ferrite** (nearly pure iron) and **pearlite** (layered ferrite and hard iron carbide, cementite); the more carbon, the more pearlite, the harder and less ductile the steel. Heat it past the **lower critical temperature A_1 (727 °C)** and the structure transforms into **austenite**, which dissolves carbon freely. What happens next depends entirely on how it cools: cool slowly and it returns to ferrite + pearlite; cool at a moderate rate and the pearlite forms finer and harder; quench it fast enough and the carbon is trapped in a distorted, extremely hard structure called **martensite**. Heat treatment is the art of choosing that cooling path. For hypoeutectoid steels (under 0.77 % C) full transformation needs heating above the upper critical line A_3; hypereutectoid steels are treated from just above A_1.

2. The four workshop treatments

PROCESS	HEATING	COOLING	RESULT AND PURPOSE
Full annealing	30–50 °C above A_3, soak	Very slow — inside the furnace	Softest state: coarse pearlite, low strength, best machinability; removes internal stress and refines cast/forged structure
Normalising	40–50 °C above A_3, soak	Still air	Finer, more uniform pearlite: stronger and slightly harder than annealed, stress relieved — the standard "as-supplied" condition
Hardening	Above A_3 (or A_1 for high-carbon),	**Quench** — brine, water, oil or air by	Martensite: maximum hardness and wear resistance, but brittle and stressed — never

	soak	severity	used untempered
Tempering	Reheat hardened steel to 150–650 °C	Air	Trades a little hardness for much toughness; the temperature chosen sets the balance (see below)

Two supporting processes belong here. **Stress-relief annealing** (550–650 °C, no phase change) relaxes welding and machining stresses without softening the steel. **Process annealing** (just below A_1) restores ductility to cold-worked stock so drawing or rolling can continue. Tempering ranges are chosen by duty: 150–250 °C keeps near-full hardness for cutting tools and dies (watch for the classic temper colours — straw ≈ 230 °C, blue ≈ 300 °C); 350–450 °C suits springs; 500–650 °C gives the tough, strong condition wanted in shafts and structural parts. Quench severity runs brine > water > oil > air: violent quenches harden deeply but risk cracking and distortion, so alloy steels — which harden more easily — are oil- or air-quenched.

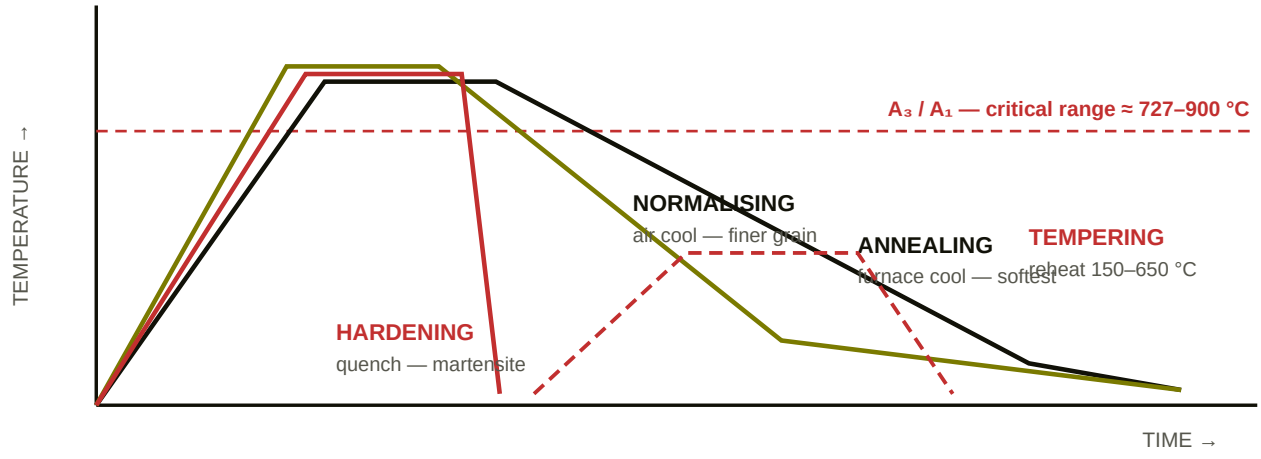


Figure 1 — Temperature–time paths of the four treatments (schematic). All start by austenitising above the critical range; the cooling path — and for tempering, the reheat — decides the final structure.

3. Case hardening — hard skin, tough core

A gear or shaft wants a wear-proof surface over a shock-absorbing core. **Carburising** gives low-carbon steel (0.1–0.2 % C) that combination: parts are held at 900–950 °C in a carbon-rich atmosphere (pack, gas or liquid salt), carbon diffuses inward, and a subsequent quench hardens only the enriched case while the low-carbon core stays tough. Case depth follows a parabolic law — depth grows with the square root of time:

$$\text{Case depth } d = k\sqrt{t}$$

WORKED EXAMPLE

A shop finds a 0.6 mm case after 2.25 h at 925 °C, so k = 0.6 / √2.25 = 0.6 / 1.5 = 0.4 mm/√h. A job needing a 1.2 mm case then takes t = (1.2 / 0.4)² = **9 hours** — doubling the depth costs four times the furnace time, which is exactly why deep cases are expensive. Alternatives: **nitriding** (nitrogen at ≈ 500 °C in ammonia — very hard, no quench, minimal distortion, but slow and only for alloy steels), **cyaniding** (carbon + nitrogen in a salt bath — quick, thin cases), and **flame or induction hardening** (heat only the surface of a medium-carbon steel and quench it — no chemistry change at all).

Part B — Moulding and casting

4. The casting route

Casting shapes metal by melting and pouring it into a mould cavity, where it solidifies. It makes shapes no other process can — engine blocks, machine beds, pump housings with internal passages — in almost any metal, at almost any size, with the cheapest tooling of the shaping processes. The route: make a **pattern** (a slightly oversized model of the part) → pack **moulding sand** around it in a two-part box (cope on top, drag below) → withdraw the pattern, leaving the cavity, and cut the **gating system** → set any **cores** (sand inserts that form holes and internal passages) → melt the metal (cupola for cast iron, crucible for small non-ferrous batches, electric arc or induction for steel and precision work) → pour, let solidify, shake out, and clean and finish the casting (fettling: cutting off gates and risers, grinding, shot-blasting).

5. Pattern allowances — why the model is not the part's size

A pattern deliberately differs from the finished part in four ways. **Shrinkage allowance**: metal contracts as it cools, so the pattern is made larger by the metal's contraction — about 10 mm/m for cast iron, 13 mm/m for aluminium, 20 mm/m for cast steel. **Draft**: vertical faces are tapered 1–3° so the pattern draws out of the sand without tearing it. **Machining allowance**: faces that will be machined carry extra metal (a few mm) to be cut away. **Rapping (shake) allowance**: the pattern is rapped to free it, slightly enlarging the cavity, so precision patterns are made a shade smaller to compensate.

WORKED EXAMPLE

A cast-steel bracket must finish 500 mm long. Pattern length = 500 + 500 × (20/1000) = 500 + 10 = **510 mm**. The same bracket in cast iron would need only 505 mm — the patternmaker's contraction rule is chosen for the metal, which is why one pattern must never be used for a different alloy.

6. Moulding sand and the gating system

Moulding sand is silica sand bonded with clay (bentonite) and a few percent of water. It must be a compromise of properties: **refractoriness** to survive the melt, **permeability** so steam and gases escape (or they blow holes in the casting), **green strength** to hold the cavity's shape, **collapsibility** so it breaks away after solidification, and enough flowability to pack around the pattern. The metal's path into the mould — the **gating system** — is engineered, not incidental: the **pouring cup** receives the stream and settles it; the tapered **sprue** carries it down (tapered so the falling stream does not tear away and suck in air); the **runner** distributes it along the mould, trapping slag; **ingates** feed the cavity quietly from below or the side; and the **riser** — a reservoir of hot metal standing over the casting — feeds liquid metal back as the casting shrinks during solidification, so the shrinkage cavity forms in the riser (which is cut off) instead of in the part. A **vent** lets mould gases out.

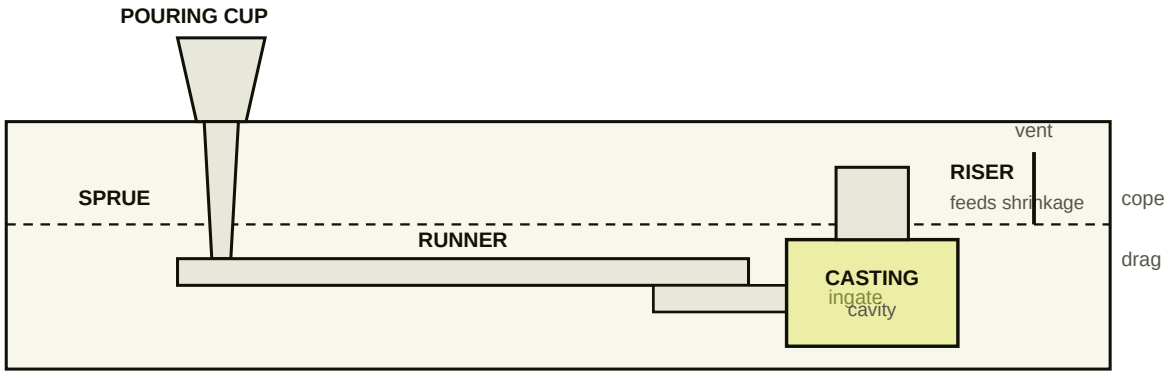


Figure 2 — A sand mould in section. The gating system delivers metal calmly to the cavity; the riser, staying liquid longest, feeds the casting's solidification shrinkage.

7. Casting defects

DEFECT	WHAT IT IS	USUAL CAUSE
Blowholes	Smooth round cavities in the metal	Gas from wet or impermeable sand, or a turbulent pour trapping air
Shrinkage cavity	Rough hollow, usually at the thickest section	No riser, or riser too small / freezing before the casting
Misrun / cold shut	Part incomplete, or two streams meet unfused	Metal too cold, poured too slowly, section too thin

Hot tear	Crack formed while the metal was still weak and hot	Mould or core resisting the casting's contraction
Sand inclusion	Sand grains embedded in the surface	Loose or weak mould face, careless closing of the mould

Alongside sand casting stand the special processes: **die casting** (metal forced into a steel die — fast, accurate, thin-walled, for aluminium and zinc parts in quantity), **investment (lost-wax) casting** (a wax pattern coated in ceramic — the finest detail and finish, for small complex parts), and **centrifugal casting** (the mould spins, flinging metal outward — dense, sound pipes and rings).

Key points

• Structure is destiny: austenite on heating; slow cool → soft pearlite, air cool → finer pearlite, quench → hard, brittle martensite.
• Anneal for softness, normalise for uniformity, harden for hardness, temper always afterwards for toughness.
• Carburising case depth grows with √time — deep cases cost disproportionate furnace time.
• A pattern is not the part: add shrinkage (CI 10, Al 13, steel 20 mm/m), draft 1–3°, and machining allowance.
• The gating system (cup → sprue → runner → ingate) delivers metal calmly; the riser feeds shrinkage so the cavity forms in the riser, not the part.
• Most casting defects trace to gas, shrinkage feeding, or pouring temperature.

Practice

1. A forged blank is hard, stressed and uneven in grain. Which single treatment both softens and refines it, and how does its cooling differ from normalising's?
2. A shaft needs a tough core and a 0.9 mm hard case. The carburising shop's constant is k = 0.45 mm/√h. How long is the carburising run?
3. Why is a hardened chisel never put into service before tempering, and what tempering range suits it?
4. Compute the pattern length for a 320 mm aluminium casting, and state the other two allowances the pattern still needs besides shrinkage.
5. A casting shows a rough hollow at its thickest boss and smooth round holes near the top face. Name both defects and one cure for each.
6. Why is the sprue tapered — narrow at the bottom — rather than straight or widening downward?

Answer key

1. **Full annealing** — heat 30–50 °C above A_3, soak, cool very slowly in the furnace: coarse soft pearlite, stress gone, grain refined. Normalising cools in still air instead — faster, giving finer, stronger pearlite and less softness.
2. $t = (d/k)^2 = (0.9 / 0.45)^2 = 2^2 =$ **4 hours**.
3. As-quenched martensite is glass-brittle and full of quench stress; the edge would chip or shatter at the first blow. Tempering at the low range, about 200–250 °C (straw colour), relieves stress and restores a little toughness while keeping nearly full hardness.
4. Aluminium shrinkage ≈ 13 mm/m: pattern = 320 + 320 × 0.013 = 320 + 4.16 ≈ **324.2 mm**. The pattern also needs **draft** (1–3° taper on vertical faces) and **machining allowance** on faces to be finished (plus rapping allowance for precision work).
5. The rough hollow is a **shrinkage cavity** — cure: a properly sized riser over the boss that stays liquid longest and feeds it. The smooth round holes are **blowholes** — cure: dry, permeable sand (correct moisture and venting) and a calmer, non-turbulent pour.
6. A falling stream accelerates and narrows (continuity: faster flow, smaller section). A sprue tapered to match that natural contraction stays full of metal; a straight or widening sprue leaves a gap around the stream, and the low pressure there sucks mould air into the metal — causing blowholes and oxidation.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib